Lab#5 – Ethernet Frames and ARP Analysis using Wireshark

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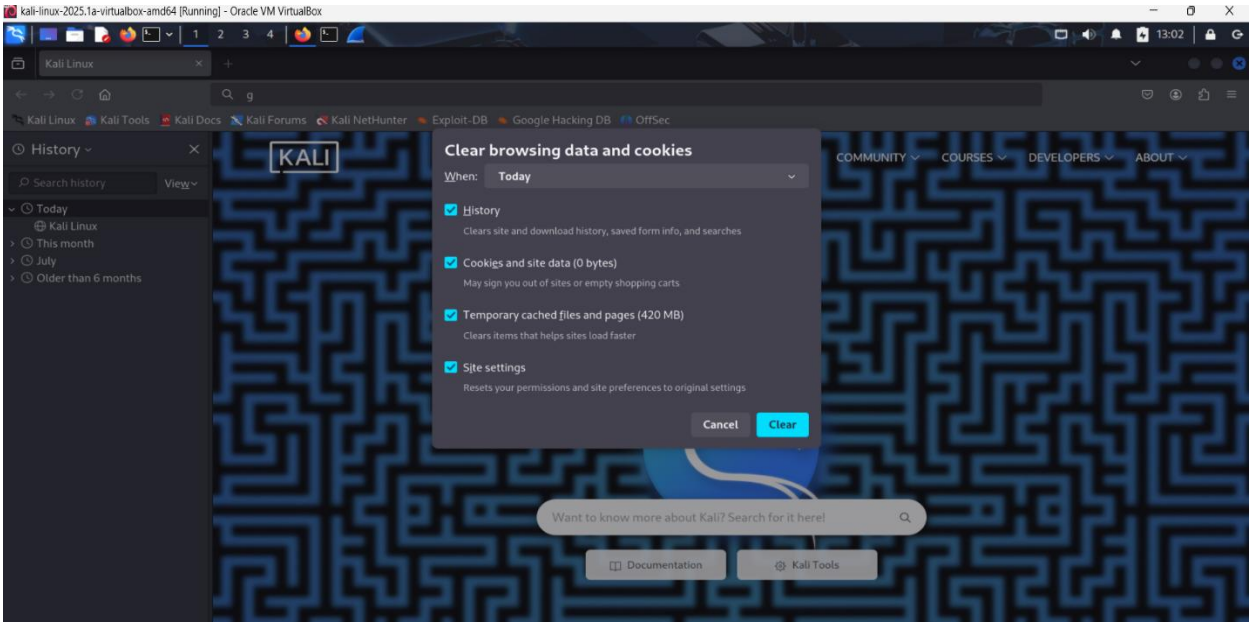
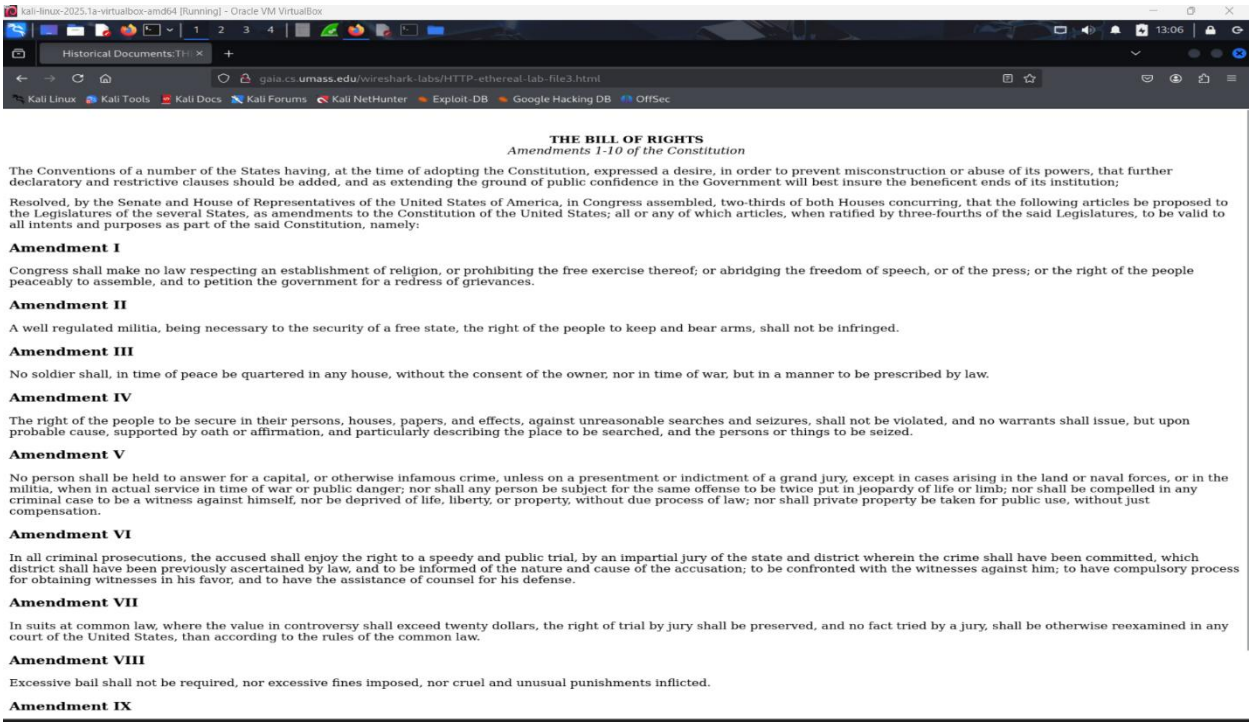
Part 1: Capturing and Analyzing Ethernet Frames

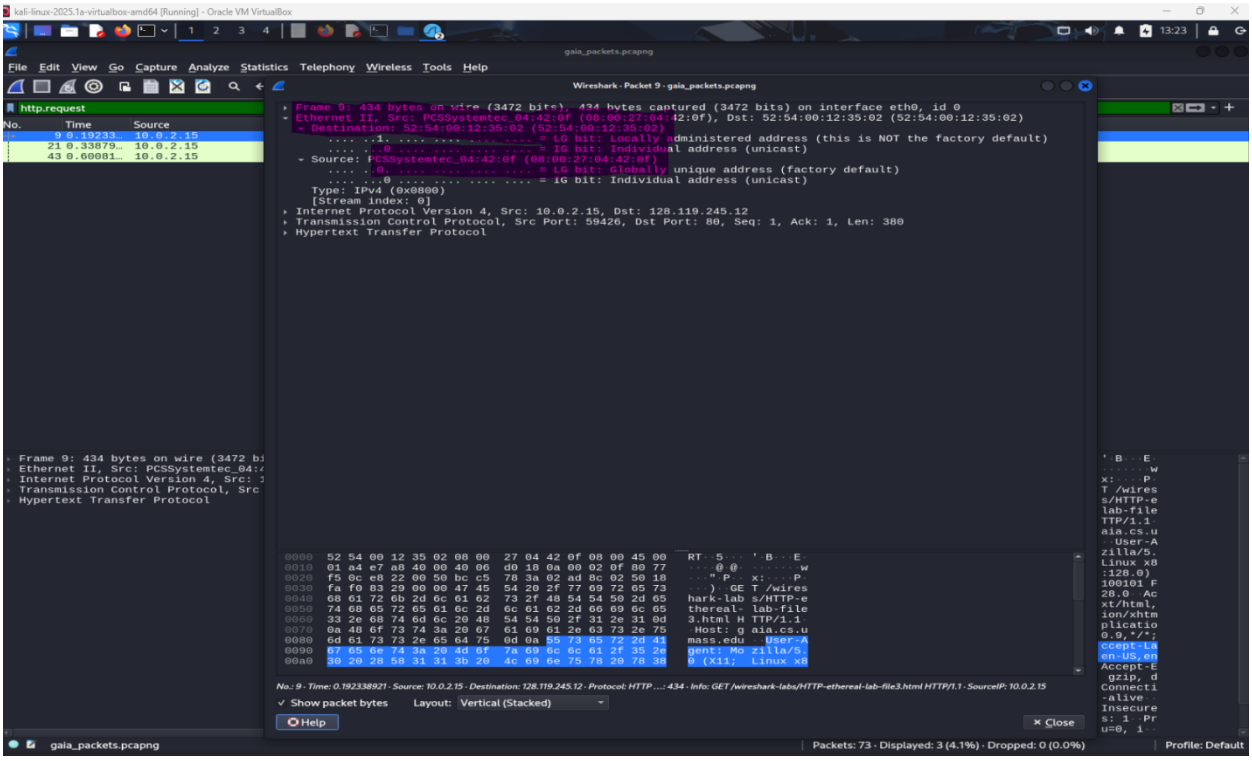
Clearing Browser Cache



Loading the given page in the browser after staring the capture



Identify Ethernet Frame Structure:



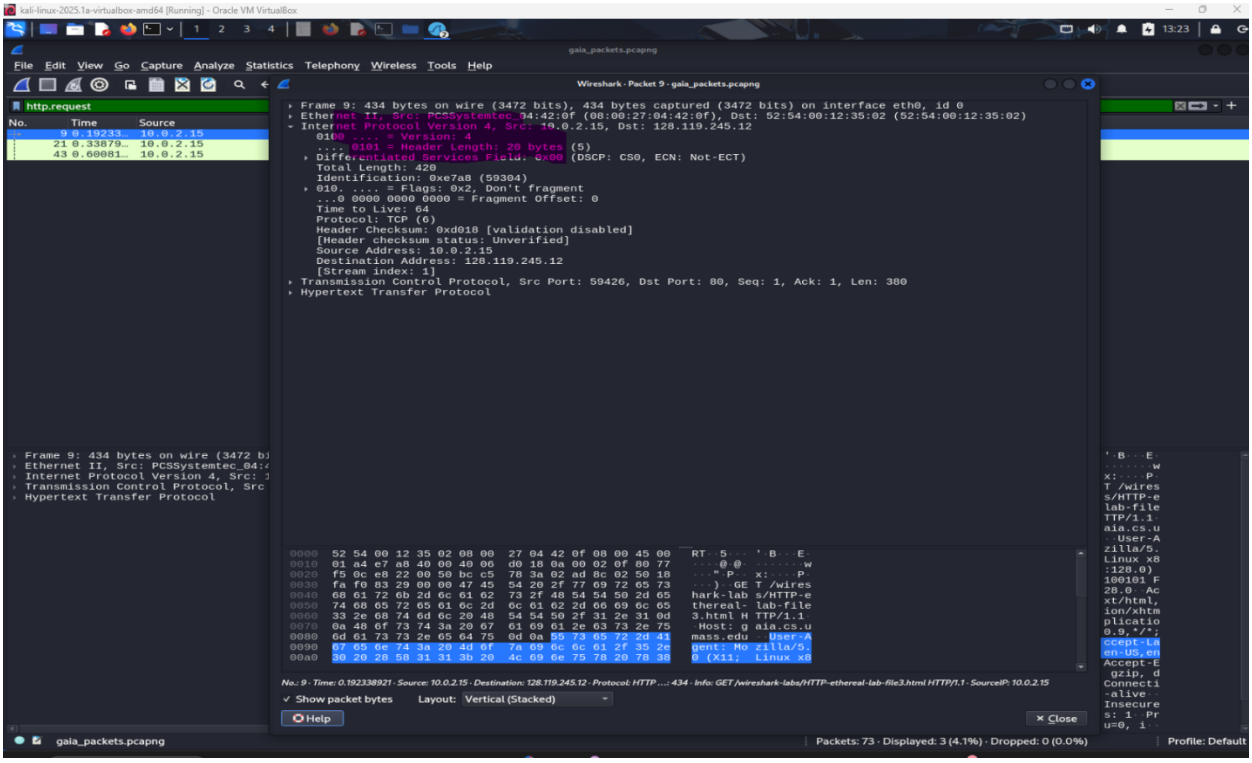
In the HTTP GET frame in screenshot the Source MAC address is PCSSystemtec_04:42:0f (08:00:27:04:42:0f) which ensures that this frame had been sent out by my local machine.

The Destination MAC address Destination: 52:54:00:12:35:02 (52:54:00:12:35:02), and this is part of the default gateway/ router.

Since Ethernet is only used within the local network segment, packets addressed to an Internet server need to be first sent to the local router, after which the packet is then forwarded via the IP routing.

The MAC of the router thus assumes the target MAC of all the off subnet traffic.

Q2 Explore Ethernet Frame Types:



Frame Number 9 Ethernet Type field = **0x0800**, which means that the contents of this Ethernet frame are an IPv4 packet.

This service is necessary as it enables the host receiving to demultiplex the frames entering it to the relevant network-layer protocol processing.

Here, the Ethernet layer passes the payload to the IPv4 protocol to process the payload.

Q3 Manufacturer Lookup for MAC Addresses

Frame 9: 434 bytes on wire (3472 bits), 434 bytes captured (3472 bits) on interface eth0, id 0

PCSSystemtec_04:42:0f

MAC: 08:00:27:04:42:0f

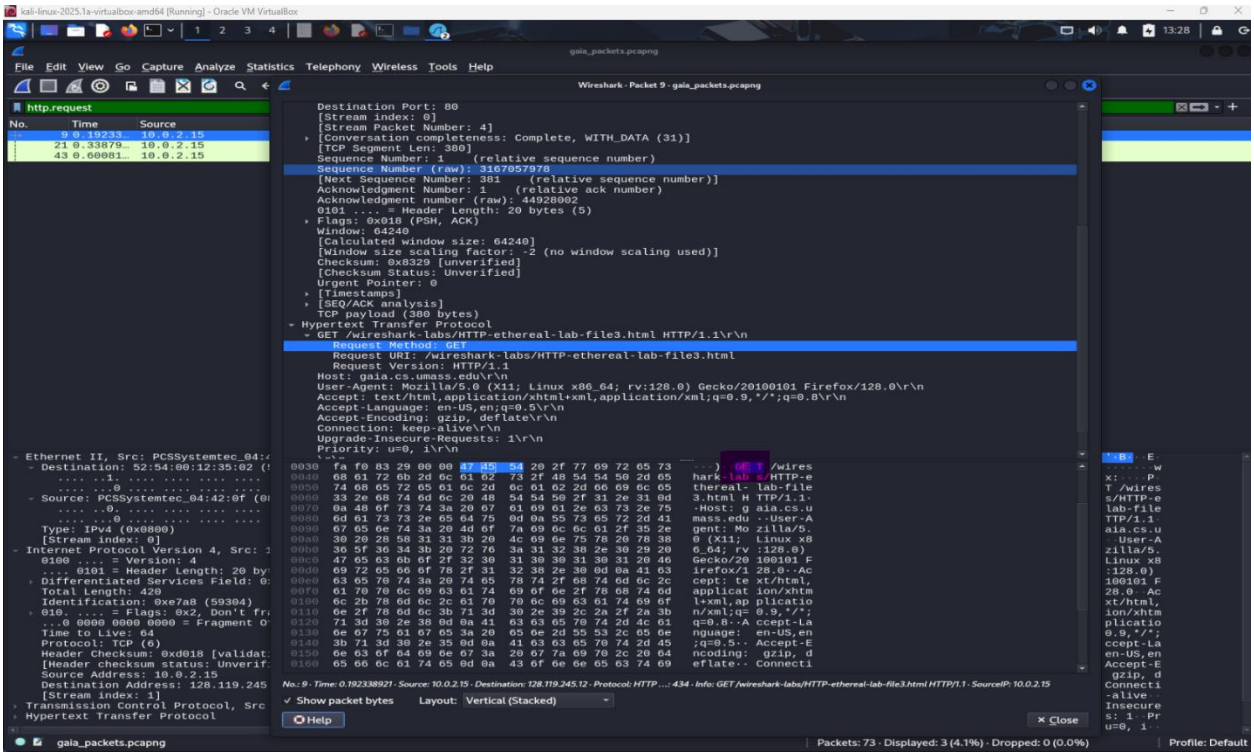
The vendor name is manufacturer PCSSystemtec above

Information which manufacturers are involved in the network is beneficial to network analysts since they can determine:

The MAC 52:54: 00: 12: 35: 02 destination address is what is normally given to a virtual NIC by the virtualization software.

The name of the MAC maker also comes in handy when troubleshooting a network and security, since it can be used to determine the type of device that is creating traffic (e.g. Intel NIC, VirtualBox adapter, router, IoT device) or identify when something is amiss in the network (e.g. spoofed or unexpected hardware).

Q4 Field Position Identification



Frame 9 Frame Solitary frame of an Intranet path, as shown in Frame 9 above, consists of a single Ethernet frame, with the ASCII code G formed at the start of the frame by the HTTP GET request:

This offset is equal to the sum of lengths of:

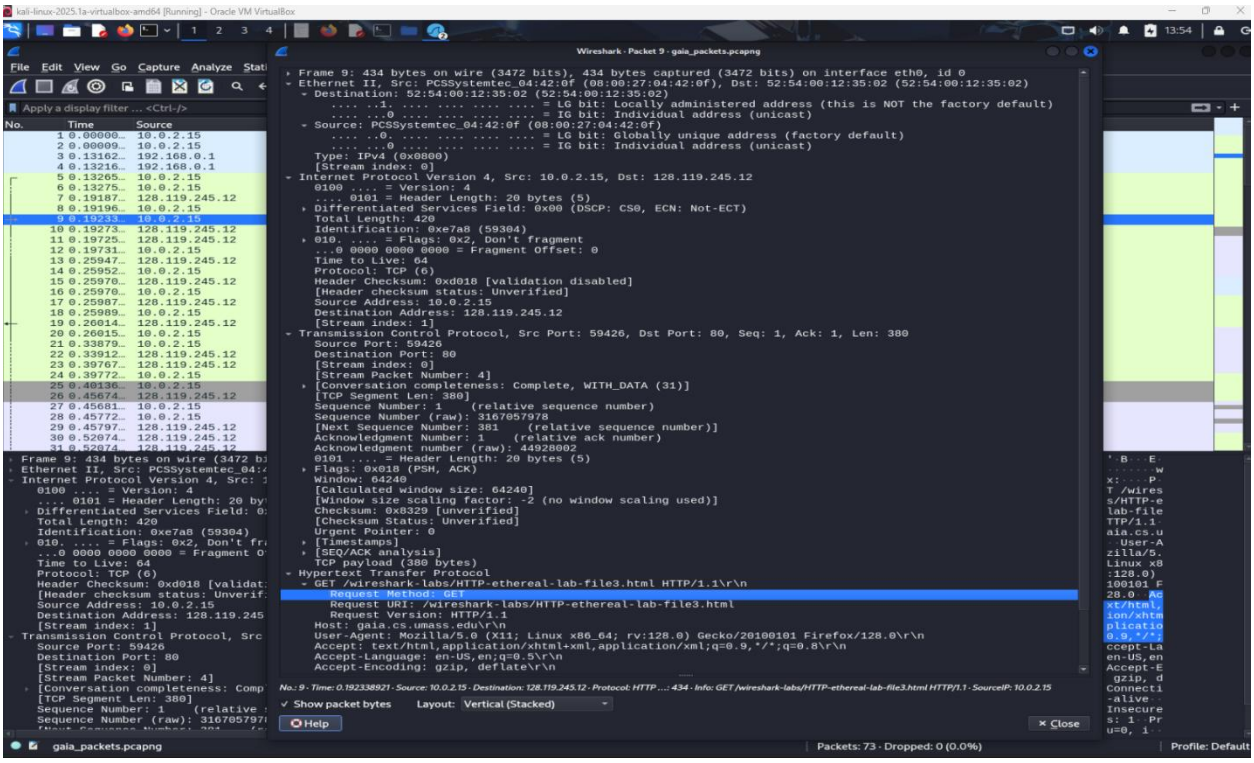
The Ethernet header (14 bytes)

The IPv4 header (typically 20 bytes, as options set it)

TCP header (usually 20 bytes, bigger if the options have been used)

The HTTP data is commenced right after the TCP header and thus the spot of the G coincides to the mark where the application-layer data begins.

Q5 Ethernet and IP Layer Encapsulation



Frame above is a clear illustration of the meaning of network-layer encapsulation:

Ethernet II (Layer 2)

Local-link addressing on the basis of the MAC addresses, IP datagram is its payload.

IPv4 (Layer 3)

Forwards logical addresses and routing. The TCP segment is its payload.

TCP (Layer 4)

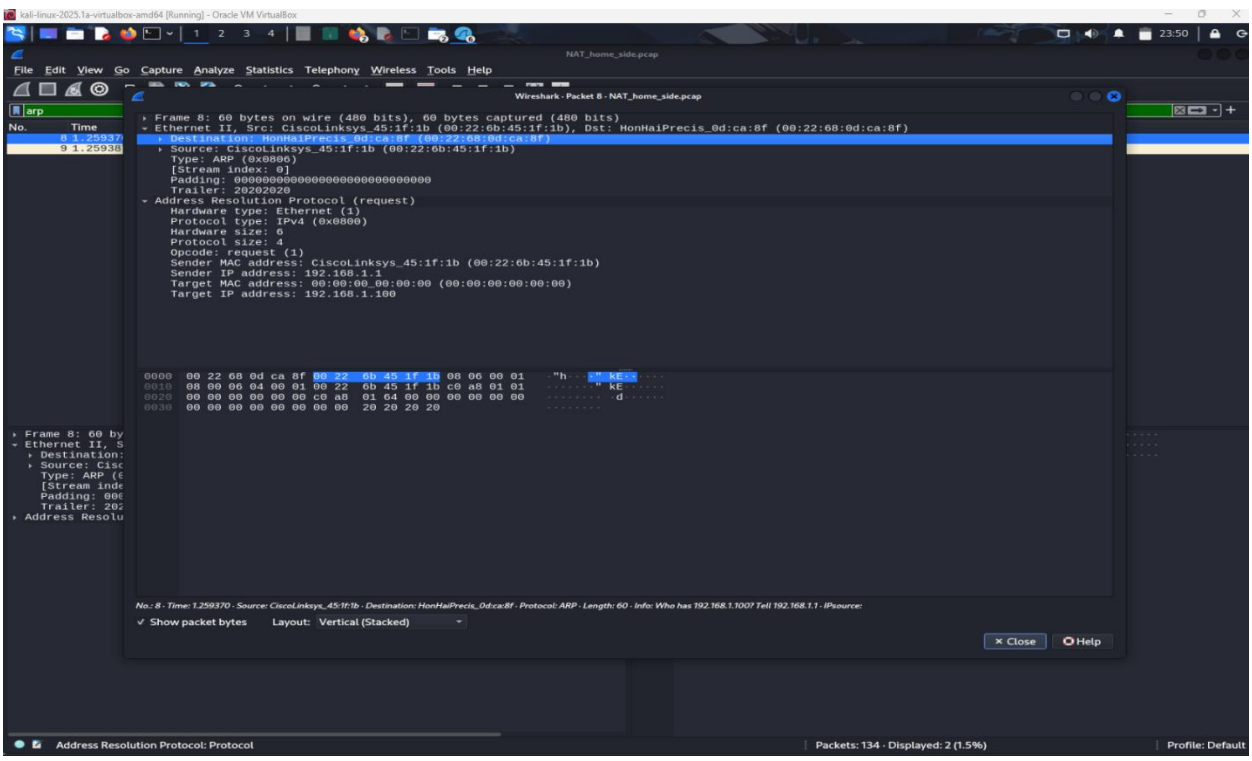
This provides reliable ordered transfer of bytes between processes networks and HTTP request is its payload.

HTTP (Layer 7)

Here it indicating the human-readable application data (GET /... HTTP/1.1").

The information of the layer above is provided with an additional header of every layer making modular architecture allows heterogeneous network interoperability retaining lower-layer mechanisms immune to application abnomility.

Q6. First ARP Packet



Frame 8 has an Ethernet II header given as

MAC Destination: HonHaiPrecis_0d:ca:8f (00:22:68:0d:ca:8f)

MAC Source: CiscoLinksys_45:1f:1b (00:22:6b:45:1f:1b)

EtherType: 0x0806 (ARP)

The EtherType value of 0x0806 is showing it has an Address Resolution Protocol (ARP) packet.

The Destination MAC is 00:22:68:0d:ca:8f and that is a **Hon Hai Precision (Foxconn**) network interface in the local LAN.

It is an ARP Request packet as indicated by ARP header:

Sender IP: 192.168.1.1

Target IP: 192.168.1.100

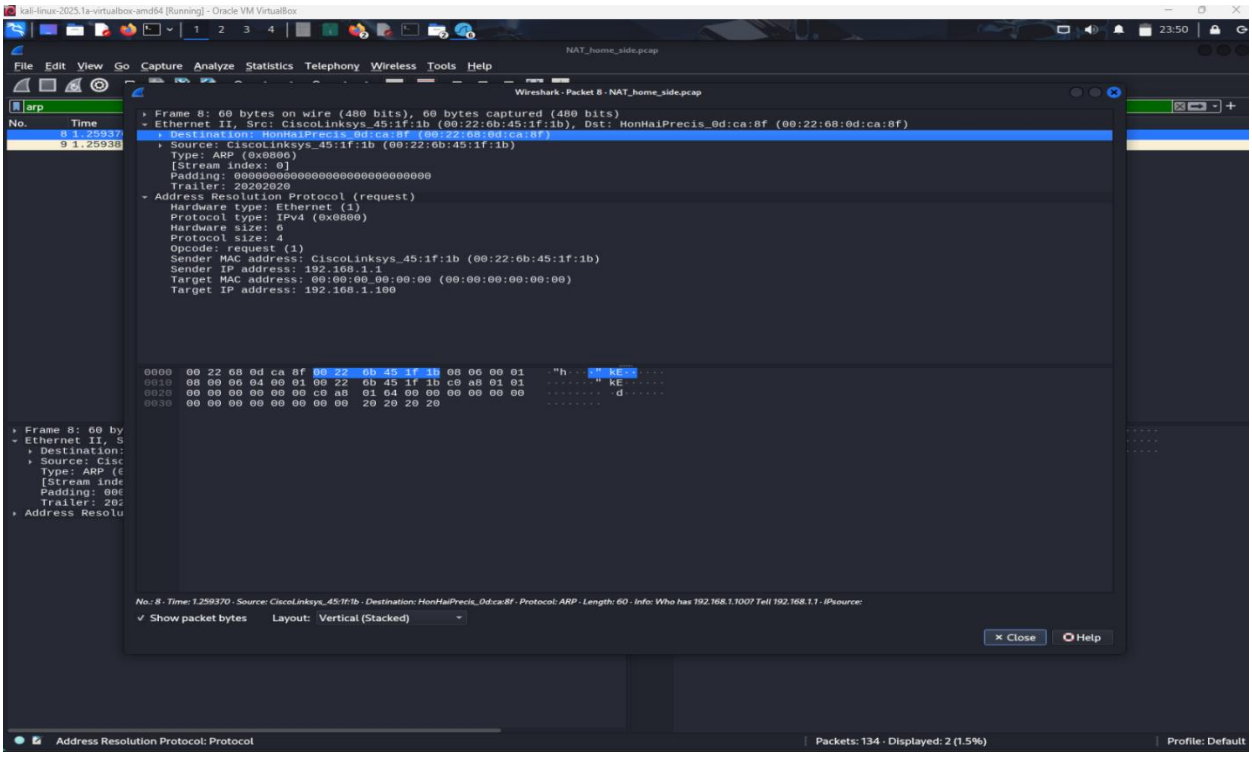
Target MAC: 00:00:00:00:00:00 (unknown)

impling that the router at 192.168.1.1 is querying:

"Who has IP 192.168.1.100? Tell me your MAC address."

The ARP message is transmitted at Layer 2 to the destination MAC address indicated in the Ethernet header (00:22:68:0d:ca:8f). Because ARP only works at the local network, the device responding to this MAC address (192.168.1.100) will respond to the request as it is expected to send its MAC address.

Q7. Explore ARP Operations



Opcode: request (1)

Sender MAC: 00:22:6b:45:1f:1b

Sender IP: 192.168.1.1

Target MAC: 00:00:00:00:00:00 (unknown)

Target IP: 192.168.1.100

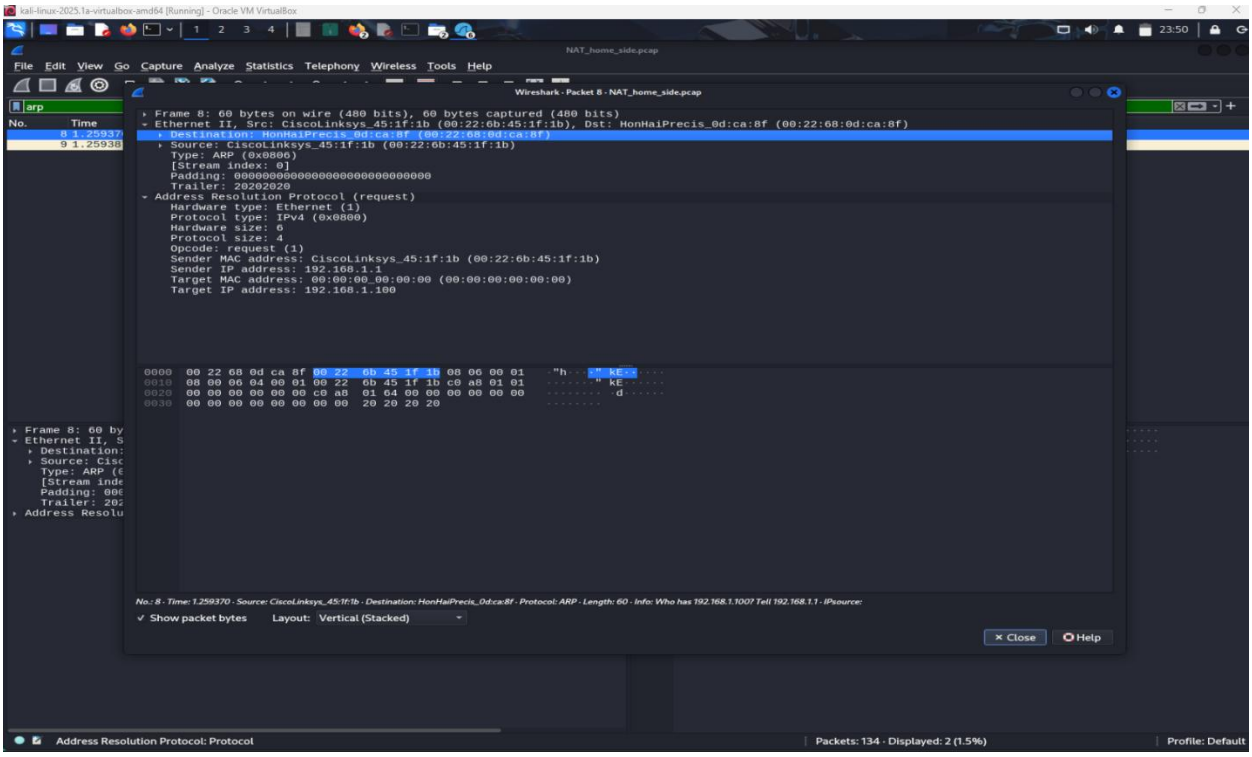
The device on the 192.168.1.1 which is the default gateway/ router using this ARP packet to learn the MAC address of the host at 192.168.1.100.

Theoretically here the packet query is asking

"I am 192.168.1.1 (MAC 00:22:6b:45:1f:1b). Who has IP 192.168.1.100? Kindly provide me with your MAC address.

The purpose of such ARP request, then, is to address the Layer-3 IP address of 192.168.1.100 into a Layer-2 MAC address, so that the router will be able to properly assemble subsequent Ethernet frames to be sent to that host on the local network.

Q8. ARP Header and Target MAC Analysis



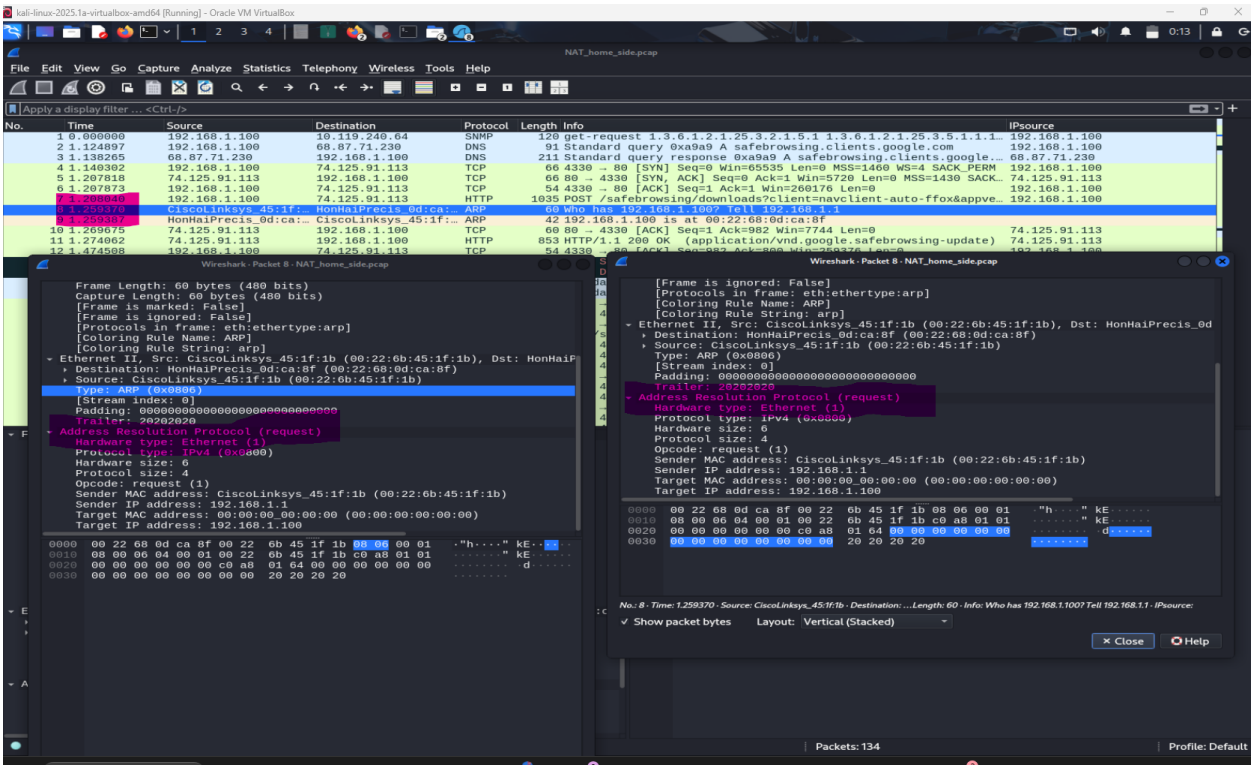
- **Ethernet Destination MAC (Layer 2):** 00:22:68:0d:ca:8f (HonHaiPrecis_0d:ca:8f)
- **ARP Target MAC (inside ARP header):** 00:00:00:00:00:00

These two fields are not identical, and there is a distinction clearly between them

The ARP Target MAC represents a 0 since the sender is not aware of the MAC address of the target IP192.168.1.100 yet. It is just that it is an ARP request that it is sending.

In Layer-2 address that this frame is being sent on the local LAN is the Ethernet Destination MAC of 00:22:68:0d:ca:8f. this is where the router is unicasting an ARP request to the last known MAC address (192.168.1.100) of 192.168.1.100 a well known common trick to check whether a previously-known mapping is still alive.

Q9 ARP Response Timing Analysis



In order to calculate the ARP round-trip time, I analyzed the ARP request packet in Frame 8 and ARP reply packet in Frame 9.

Frame 8 is 1.259370 seconds Category and ARP frame 9 is 1.259387 seconds.

The RTT is therefore:

1.259387-1.259370 = 0.000017 seconds

which equals 0.017 milliseconds.

Such a very small RTT should be expected when asking ARP to exchange in a local layer since ARP is Layer 2 and it does not go through routers and other external networks. It indicates the average differing latency of communicating to another host on the identical segment of LAN.

Q10 MAC Address and ARP Spoofing Simulation

MAC and ARP spoofing has a direct impact on the way Wireshark decodes the communication over the link-layer since Wireshark uses the Ethernet source and ARP sender fields to determine the hosts.

When a host willingly alters its MAC address, or sends spoofed ARP responses purporting to be the owner of an IP address, the ARP table in other computers will contain wrong IP-MAC entries.

Spoofing will be presented as: in Wireshark.

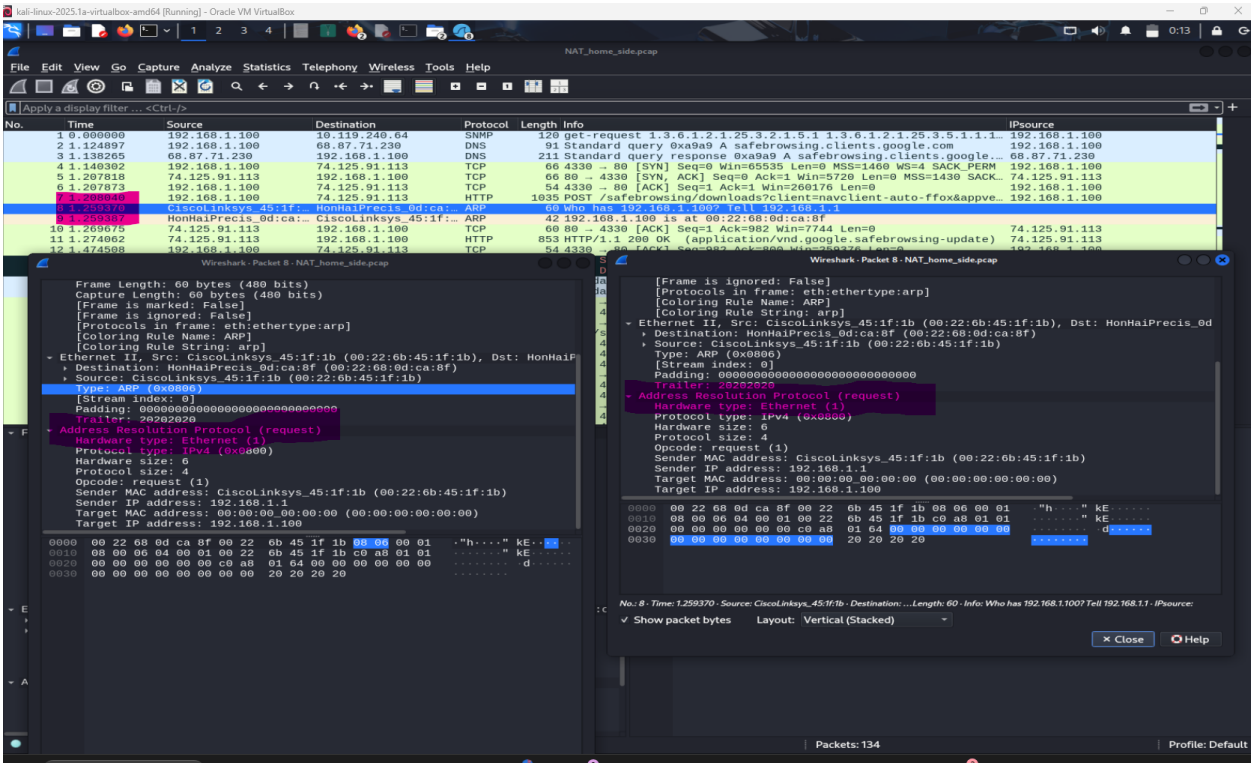
ARP response messages with non-matching MAC address with the host.

Unsolicited ARP responses (ARP gratuitous packets or ARP poison packets).

IP packets that come out to have a MAC address not belonging to the vendor or device expected.

Such manipulations may result in man in the middle attacks, traffic redirection attacks and network integrity loss. Wigenshark will indicate seemingly valid packets, though the hardware identities beneath will not be true since the attacker has hacked the ARP cache tables.

Q11 Dynamic ARP Cache Observation



ARP caching will help greatly to minimize the amount of ARP requests that have been sent across a local network. There are no additional ARP requests with this IP address indicated in the packet trace after Frame 9 that has the mapping 192.168.1.100 - 00:22:68:0d:ca:8f.

This action proves the role of the ARP cache:

after one of the hosts manages to solve an IP-to-MAC mapping, it adds the record, which lasts a certain duration (usually 1-20 minutes on a system basis). This means that the related packets can be sent immediately and no new ARP manual is issued.

There are only one ARP request (Frame 8) and one ARP reply (Frame9) in this trace. Subsequent frames are unicast transmission with the host MAC address that has been already cached. This abatement of broadcast traffic enhances the network efficiency and reduction of latency of the subsequent connections.